A correlated prior on galaxy bias across thin redshift bins

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1 Introduction: why an intermediate bias model

A Fisher forecast of galaxy clustering in thin redshift bins must decide how to parametrize the galaxy bias of each bin. The bins belong to a few broad groups: the DESI tracer samples and their chunks, such as BGS, LRG1, LRG2, LRG3, ELG1 and ELG2. Within one group the galaxies are selected the same way, so their biases are expected to be similar but not identical.

The default model ties all bins of a group to a single amplitude. If $D(z)$ is the linear growth factor, normalized to $D(0) = 1$, the tied model sets

$$b_i \;=\; \frac{B_g}{D(z_i)}, \tag{1}$$

where b_i is the linear bias of bin i, z_i is the bin's redshift, and B_g is one free amplitude per group g. The $1/D$ evolution keeps the clustering amplitude $b(z)\,D(z)$ of the group constant, the behavior measured for DESI-like luminous red galaxies [6]. It is not the law of a fixed set of galaxies that follow the flow of matter ("passive evolution"); Section 3.1 derives that law and the small difference between the two.

The opposite extreme frees one bias per bin, with flat priors and no connection between neighbors ("independent"). The dark-energy figure of merit of the forecast depends on which extreme is chosen, and the question is how much of it rests on the tying itself. To answer that, this note builds the intermediate model: bins may deviate from the group's trend, but neighboring bins must deviate *together*.

2 The models

2.1 Tied, independent, and correlated biases

The model multiplies the tied bias by a fractional deviation δ_i,

$$b_i \;=\; \frac{B_g}{D(z_i)}\,\big(1+\delta_i\big), \tag{2}$$

with B_g still free (flat prior). The deviations receive a Gaussian prior that is correlated inside a group,

$$\big\langle \delta_i\,\delta_j \big\rangle \;=\; \sigma_b^2\,\exp\!\left(-\,\frac{|z_i - z_j|}{L}\right), \tag{3}$$

and zero across groups. Here σ_b is the prior standard deviation of a single bin's fractional deviation, and L is the correlation length in redshift: bins much closer than L deviate almost as one, bins much farther apart deviate independently.

In the Fisher matrix the convenient parameters are the per-bin amplitudes

$$\beta_i \;\equiv\; B_g\,\big(1+\delta_i\big), \qquad \text{fiducial } \beta_i = B_g, \tag{4}$$

so that $b_i = \beta_i / D(z_i)$. Since $\beta_i - B_g = B_g\,\delta_i$, the prior covariance of the β_i within one group is

$$\mathsf{C} \;=\; s^2\,\mathsf{R}, \qquad s \;=\; \sigma_b\,B_g, \qquad \mathsf{R}_{ij} \;=\; \exp\!\left(-\,\frac{|z_i - z_j|}{L}\right). \tag{5}$$

The spread s carries the units of a bias amplitude; the correlation matrix R is dimensionless, with ones on the diagonal.

2.2 What σ_b and L mean

The kernel of Eq. (3) has two parameters, and they control two different things. The diagonal gives the wander of a single bin,

$$\mathrm{Var}(\delta_i) \;=\; \sigma_b^2, \tag{6}$$

so σ_b sets how far any one bin may drift from the group's trend $B_g/D(z)$. The correlation coefficient r of two bins i and j follows from the same kernel. By definition,

$$r \;=\; \frac{\langle \delta_i \delta_j \rangle}{\sqrt{\langle \delta_i^2 \rangle\,\langle \delta_j^2 \rangle}} \;=\; \frac{\sigma_b^2\, e^{-\Delta z/L}}{\sigma_b^2} \;=\; e^{-\Delta z/L}, \tag{7}$$

with $\Delta z = |z_i - z_j|$ the bin separation. The amplitude σ_b cancels: the correlation is set by the separation and L alone.

The jump between two bins now follows from the two pieces above: with $\mathrm{Var}(\delta_i) = \sigma_b^2$ and $\mathrm{Cov}(\delta_i, \delta_j) = \sigma_b^2\, e^{-\Delta z/L}$, the variance of the difference is $\mathrm{Var}(\delta_i) + \mathrm{Var}(\delta_j) - 2\,\mathrm{Cov}(\delta_i, \delta_j)$:

$$\mathrm{Var}(\delta_i - \delta_j) \;=\; 2\,\sigma_b^2\left(1 - e^{-\Delta z/L}\right). \tag{8}$$

For separations much smaller than the correlation length, $\Delta z \ll L$, the exponential expands as $e^{-x} \approx 1 - x$, and

$$\mathrm{Var}(\delta_i - \delta_j) \;\approx\; \frac{2\,\sigma_b^2\,\Delta z}{L}, \qquad \text{rms jump} \;\approx\; \sigma_b\sqrt{\frac{2\,\Delta z}{L}}. \tag{9}$$

So L sets the correlation between bins, σ_b sets the wander of each bin, and *together* they set the jump a thin bin may take relative to its neighbor.

The combination that controls small separations is the roughness

$$\kappa \;\equiv\; \frac{\sigma_b}{\sqrt{L}}, \qquad \mathrm{Var}(\delta_i - \delta_j) \;\approx\; 2\,\kappa^2\,\Delta z. \tag{10}$$

The second form is the signature of diffusion: the variance of the increment grows linearly with the separation, exactly as the distance walked by a random walker grows linearly in time. On scales well below L the deviation curve $\delta(z)$ therefore looks like a random walk with diffusion coefficient κ^2; only on scales beyond L does the finite wander σ_b pull it back toward zero.

Worked example. Take neighboring thin bins with $\Delta z = 0.02$ and correlation length $L = 0.05$. Then $\Delta z/L = 0.4$ and

$$r \;=\; e^{-0.4} \;=\; 0.670. \tag{11}$$

The exact rms jump from Eq. (8) is $\sigma_b\sqrt{2\,(1-0.670)} = 0.81\,\sigma_b$: for $\sigma_b = 10\%$ the neighboring bins may differ by 8.1% (one sigma), and for $\sigma_b = 1\%$ by 0.81%. The Taylor form of Eq. (9) gives instead $\sigma_b\sqrt{0.8} \;=\; 0.89\,\sigma_b$, an 8.9% jump. The approximation overestimates because $\Delta z/L = 0.4$ is not actually small: the expansion replaces $1 - e^{-x}$ by x, and x exceeds $1 - e^{-x}$ already at second order ($x^2/2$), by about ten percent here.

2.3 The free group amplitude and the projection

A plain Gaussian prior with covariance C would penalize moving all β_i of a group up or down together, but that uniform shift is exactly the free amplitude B_g, which the tied model never constrains. Keeping the penalty would make every comparison against the tied model unfair. The repair removes the uniform direction from the prior: with $\mathbf{1} = (1, \ldots, 1)^{\mathsf{T}}$ the ones vector with n entries (n the number of bins in the group), the prior precision of a group's amplitudes is the projection

$$\mathsf{P} \;=\; \mathsf{C}^{-1} \;-\; \frac{\mathsf{C}^{-1}\mathbf{1}\,\mathbf{1}^{\mathsf{T}}\mathsf{C}^{-1}}{\mathbf{1}^{\mathsf{T}}\mathsf{C}^{-1}\mathbf{1}}. \tag{12}$$

Four statements, proven in the appendices, carry the construction. $\mathsf{P}\,\mathbf{1} = 0$: a uniform shift of the whole group costs nothing, so B_g stays as free as in the tied model. A pure shape change keeps exactly the penalty the plain prior gives it. P is positive semidefinite with rank $n - 1$: of a group's n directions, one is free and the other $n - 1$ are charged. And P is precisely the prior obtained by sampling B_g with a flat prior and marginalizing it. Appendix A computes P for two bins, Appendix B derives these statements three ways, and Appendix C treats n bins, including the tied and independent limits.

Across groups the prior of Eq. (3) is zero, so the full prior precision is block diagonal: one projected block per group, each group with its own free amplitude. How the blocks enter the Fisher matrix is Appendix G.

3 Which L and σ_b are physical

Which deviations from the tied trend does known galaxy-bias physics produce? One case of bias evolution can be solved in closed form: a population whose members are conserved. This section derives that case, measures its deviation from the tied trend, compares the deviation with what the prior permits, weighs the published measurements, and ends at the scanned grid and its plausible region.

3.1 A conserved population: the bias continuity equation

Take a sample whose membership never changes: no galaxy enters, none leaves. Assume the galaxies move with the matter on large scales (no velocity bias), and work in linear theory.

Under these assumptions the matter overdensity δ_m (the fractional fluctuation of the matter density) and the galaxy-count overdensity δ_g obey the same linear continuity equation, with the same peculiar velocity field on the right-hand side: number conservation gives each left-hand side, and the common motion gives the shared right-hand side. Subtracting the two equations,

the excess clumping $\delta_g - \delta_m$ is frozen: whatever clumping difference the sample is born with never changes. Appendix E spells out every step.

Let z_c be a reference redshift where the sample has linear bias b_c (linear bias means $\delta_g = b\,\delta_m$), and let the matter grow as $\delta_m \propto D$. Carrying the frozen difference from z_c to any other redshift gives [1]

$$b(z) - 1 \;=\; \big(b_c - 1\big)\,\frac{D(z_c)}{D(z)}. \tag{13}$$

The excess clumping set when the galaxies formed stays fixed while the matter clustering grows underneath it, so the bias decays toward one with time. Tegmark and Peebles [2] extend the calculation to stochastic bias, where newly formed galaxies trace the matter imperfectly, and find the same decay of b toward one; Appendix E summarizes their result.

The tied shape $b = B/D$ freezes a different quantity. Constant $b\,D$ means $\delta_g = b\,\delta_m \propto (B/D)\,D = B$: the galaxy clustering amplitude itself does not grow. A conserved population never does that; its δ_g keeps growing through the matter term in Eq. (50). Constant $b\,D$ instead describes a selection: at each redshift it picks a different and rarer set of objects, whose stronger clumping compensates the smaller D.

3.2 The deviation from the tied trend

Place the reference at a group's midpoint, z_c, with $b_c = B/D(z_c)$ from the group's fiducial amplitude B. The tied bias is $b_{\rm tied}(z) = b_c\,D(z_c)/D(z)$ and the conserved bias is Eq. (13). Their ratio defines the deviation $\delta(z)$ of Eq. (2), $b = b_{\rm tied}(1+\delta)$:

$$1 + \delta(z) \;=\; \frac{1 + (b_c - 1)\,D(z_c)/D(z)}{b_c\,D(z_c)/D(z)} \;=\; \frac{D(z)/D(z_c) + b_c - 1}{b_c}, \qquad \delta(z) \;=\; \frac{D(z)/D(z_c) - 1}{b_c}. \tag{14}$$

At the midpoint $\delta = 0$. At lower redshift the matter has grown (D larger than $D(z_c)$), so $\delta > 0$; at higher redshift $\delta < 0$. Because $D(z)$ is smooth, $\delta(z)$ is nearly linear in z across a group.

Table 1 evaluates Eq. (14) in flat ΛCDM with $\Omega_m = 0.3$, using the project's fiducial amplitudes. In the language of the prior, a conserved population is a smooth tilt: coherent over the whole group, so L longer than the group width, with an amplitude of a few percent at the group ends and a step of 0.3–0.7% between neighbors at $\Delta z = 0.02$.

Group	B	z range	b_c	δ at the group ends	per $\Delta z = 0.05$
BGS	1.34	0.1–0.4	1.5	+5.3%, −4.9%	1.7%
LRG1	1.70	0.4–0.6	2.2	+2.3%, −2.2%	1.1%
LRG3	1.70	0.8–1.1	2.7	+2.6%, −2.3%	0.8%

Table 1: The conserved-population deviation $\delta(z)$ of Eq. (14) from the tied trend, for flat ΛCDM with $\Omega_m = 0.3$, the reference z_c at the group midpoint, and $b_c = B/D(z_c)$ from the project fiducials. The last column is the change of δ across one $\Delta z = 0.05$ step at the midpoint.

3.3 The prior's yardstick: the neighbor rms

The prior's quantity to compare with this tilt is the rms change between neighbors of Eq. (8). Table 2 evaluates $\sigma_b\sqrt{2\,(1 - e^{-\Delta z/L})}$ at several (σ_b, L) pairs and for the conserved population

of Table 1.

Case	over $\Delta z = 0.02$	over $\Delta z = 0.05$
$\sigma_b = 3\%$, $L = 0.05$	2.4%	3.4%
$\sigma_b = 1\%$, $L = 0.1$	0.6%	0.9%
$\sigma_b = 5\%$, $L = 0.02$	5.6%	6.8%
$\sigma_b = 10\%$, $L = 0.05$	8.1%	11.2%
$\sigma_b = 20\%$, $L = 0.02$	22.5%	27.1%
conserved population	0.3–0.7%	0.8–1.7%

Table 2: The rms change of the bias deviation between two bins, $\sigma_b\sqrt{2\,(1 - e^{-\Delta z/L})}$, at several (σ_b, L) pairs, and for the drift of a conserved population (the per-group range of Table 1).

At $\sigma_b = 1\%$ with $L = 0.1$ the prior allows neighbor differences of 0.6% over $\Delta z = 0.02$, matching the conserved drift. At $\sigma_b = 3\%$ with $L = 0.05$ the allowance is several times that drift, and at $\sigma_b = 10\%$ or 20% it reaches 8–27%, beyond any bias evolution identified in this note.

3.4 What breaks conservation in a selected sample

Four mechanisms break the conserved-population assumptions.

Threshold drift. The target selection holds a roughly constant comoving density until the flux limit bites; past that redshift the sample keeps only its brighter members, which sit in more massive halos and are more biased. The resulting bias change is smooth in redshift, with L comparable to the group width.

Color-selection features. The selection cuts in color space can respond quickly in redshift when a spectral feature crosses a filter edge. The sharpest common feature is the 4000 Å break: a filter edge 100–200 Å wide maps to $\Delta z \approx 0.03$–0.05 at that break, and photometric scatter smears the transition further. Its amplitude in bias is unknown.

Mergers. Two members that merge become one, so the number of galaxies is not conserved even before any selection acts. The size of this effect on these samples is unknown.

Emission-line selection. An ELG enters the sample while its emission lines are strong, so the membership follows the star-formation phase of each galaxy rather than one conserved set of objects.

How much each of these moves the bias, and over which Δz, is not known from theory alone; it is a question about the measured selection function of each sample.

3.5 What the measurements say

The mechanisms above can be weighed against published measurements. Each statement below carries a tag: *measured* means it is read from the cited source; *derived* means it is computed here from measured inputs; *assumed* means it is a judgment not yet backed by a measurement. Equation (13) itself is from Fry [1], which has no arXiv record; Tegmark and Peebles [2] (section

"After the galaxy formation epoch") rederive it. Appendix F summarizes each paper below: the sample, the method and the numbers.

Massive red galaxies follow the conserved shape to the precision of the published tests. The passive-evolution analyses find merger corrections of a few percent per Gyr (measured [3, 4, 5]), and the DESI-like LRG bias of [6] stays within $\pm 2.3\%$ of the constant clustering amplitude $b = 1.5/D$ at a bin spacing of $\Delta z \approx 0.1$ below $z = 0.83$, with the highest bin $6.4 \pm 6.4\%$ high (measured biases; residuals derived here).

The selections drift. The LRG cuts hold the comoving density constant at $0.4 < z < 0.8$ and hand over to a fixed flux limit beyond (measured [7]); through the threshold-bias relation of [12] (measured), that flux limit makes the LRG3 bias drift by 6–11% per $\Delta z = 0.1$ on top of the B/D shape (derived), matching the low end of the measured highest-bin offset. The magnitude-limited BGS (measured limit, [13]) drifts by 10–25% across its group through the same mechanism (derived).

The ELG bias shape is stable: HOD fits in two bins covering $0.8 < z < 1.6$ agree within one sigma (measured [15]). The one short-separation mechanism found in any tracer is the set of redshift windows where the [OII] 3727 Å doublet, which carries the ELG redshift, lands on sky emission lines and loses signal-to-noise, especially at $z > 1.5$ (mechanism measured [14]): each window is $\Delta z \approx 0.001$–0.01 wide, and its amplitude in bias is assumed, of order 1–2%.

No measurement resolves bias below $\Delta z \approx 0.1$. The finest binning is the $\Delta z \approx 0.1$ of [6]; [15] use 0.3 and 0.5; and the lensing tomography of the same samples [9, 10, 11], built on the photometric samples of [8], uses about 0.15 with one free bias per bin. Below $\Delta z \approx 0.1$ the thin-bin statements of this note rest on the continuity equation and on the selection arguments.

3.6 Drift versus loss of memory between bins

Every deviation found above, except the sky-line windows, is a monotonic drift: the conserved-population tilt, the LRG3 magnitude-limit drift and the BGS flux-limit drift each move in one direction across their group. A monotonic drift is a coherent, long-correlation deformation, a tilt; it is not bin-to-bin decorrelation, so it does not argue for a short L. Within the exponential kernel this note analyzes, the measured drifts point to the long end of the scanned L grid, and only the sky-line windows point to the short end.

3.7 The scan and the plausible region

The forecast scans

$$L \in \{0.02,\ 0.05,\ 0.1,\ 0.2\} \quad \text{(in redshift)}, \qquad \sigma_b \in \{1\%,\ 3\%,\ 5\%,\ 7\%,\ 10\%\}, \tag{15}$$

with baseline $L = 0.05$ and $\sigma_b = 3\%$ (a provisional choice: the measurements of Section 3.5 can still move it), and the plausible region $L = 0.02$–0.1, $\sigma_b = 1$–5%. The upper end of L is physical, not numerical: a sample's flux limit and color cuts change which galaxies enter the sample as redshift grows, and that change of galaxy mix happens over $\Delta z \sim 0.1$, so bias deviations should not be assumed coherent over separations much beyond that: $L \lesssim 0.1$.

In Table 2, the baseline allows neighbor differences of 2.4% over $\Delta z = 0.02$, several times the conserved drift; the gentlest corner of the plausible region ($\sigma_b = 1\%$, $L = 0.1$) matches that drift, and the sharpest ($\sigma_b = 5\%$, $L = 0.02$) allows roughly eight times it. The rows $\sigma_b = 10\%$ and 20% show what the conserved-population comparison rules out: neighbor differences of 8–27%. Both sit above the plausible region, and $\sigma_b = 20\%$ is outside the scanned grid.

Shape freedom costs dark-energy information: the Fisher matrix spends part of the data constraining the bin biases instead of w_0 and w_a; Section 4 measures the cost across the scanned grid. Relative to a conserved population, the figure of merit computed anywhere in the plausible region except its gentlest corner is pessimistic.

This comparison is not a proof, because a DESI sample is not conserved: the selection, not gravity, decides which galaxies enter each bin.

4 Results

The forecast behind the scan is the FoM(w_0, w_a) of the DESI clustering band powers alone (C_ℓ^{gg}, no shear), computed non-Limber with the ultraviolet-fraction Fourier scale cut at $k_{\max} = 0.075\,h/\mathrm{Mpc}$ of [16], which follows DeRose et al. 2025. The figure of merit is the usual inverse area of the (w_0, w_a) error ellipse, $\mathrm{FoM} = [\det \mathrm{Cov}(w_0, w_a)]^{-1/2}$: larger is better. Four thin-bin layouts are scanned: the $\Delta d_{\mathrm{com}} = 62$ Mpc layout (bins of equal comoving width) and the fixed-width layouts with $\Delta z = 0.02$, 0.05 and 0.2. In each layout the grid of Eq. (15) is bracketed by the tied and independent models.

Preliminary: linear bias only. The forecast models the galaxy field with linear bias alone. The absolute FoM values will move when higher-order bias terms enter the model; the comparisons between bias priors on a fixed layout are the robust content of this section.

Figure 1 places every scanned (L, σ_b) between its layout's two caps. At the baseline ($\sigma_b = 3\%$, $L = 0.05$) the four layouts give FoM = 11.8, 8.05, 2.01 and 0.779, which is 0.36, 0.40, 0.24 and 0.41 of the respective tied values. At $\sigma_b = 1\%$ the tied fraction spans 0.45–0.79 across layouts and correlation lengths. At $\sigma_b \geq 5\%$ the $\Delta d_{\mathrm{com}} = 62$ Mpc and $\Delta z = 0.05$ layouts sit near their independent values: most of the information the tying provided is already spent. One scanned point exceeds the LSST Y1 3x2pt reference of 17.3 (the dashed line): the $\Delta d_{\mathrm{com}} = 62$ Mpc layout at $\sigma_b = 1\%$, $L = 0.2$, with FoM = 20.8.

Figure 2 resolves the same numbers on the (L, σ_b) grid. $\Delta z = 0.02$ is the only layout where L matters strongly: at $\sigma_b = 3\%$ its FoM runs from 6.8 at $L = 0.02$ to 10.9 at $L = 0.2$. The reason is the short neighbor separation: with $\Delta z = 0.02$ the neighbor correlation $e^{-\Delta z/L}$ stays high over the whole L grid, from 0.37 at $L = 0.02$ to 0.90 at $L = 0.2$, so lengthening L keeps converting bin-to-bin freedom back into effective tying. In the other layouts the bins are as far apart as L itself or farther, the neighbor correlation is low everywhere on the grid, and L moves the FoM only weakly.

Figure 3 tests whether one number controls the cost. Its horizontal axis is $\sqrt{2}\,\kappa = \sigma_b\sqrt{2/L}$, the rms neighbor change per unit $\sqrt{\Delta z}$ of Eq. (10). The points do not collapse onto a single curve per layout. The reason is that the roughness controls only the short-separation jumps: at fixed κ, a longer L also carries a larger wander $\sigma_b = \kappa\sqrt{L}$, so the bias can drift coherently over the whole group, a freedom the roughness alone does not capture. The Fisher matrix pays for both kinds of freedom, so no single-variable description of the grid should be expected.

The scan reaches its two limits numerically, as Appendix C.4 says it must. At $\sigma_b = 0.01\%$ with $L = 0.05$ the four layouts give FoM = 32.93, 19.93, 8.285 and 1.907, against tied values 32.94, 19.93, 8.288 and 1.908; at $\sigma_b = 1000\%$ they land on the independent values 10.69, 4.016, 1.225 and 0.5154 exactly.

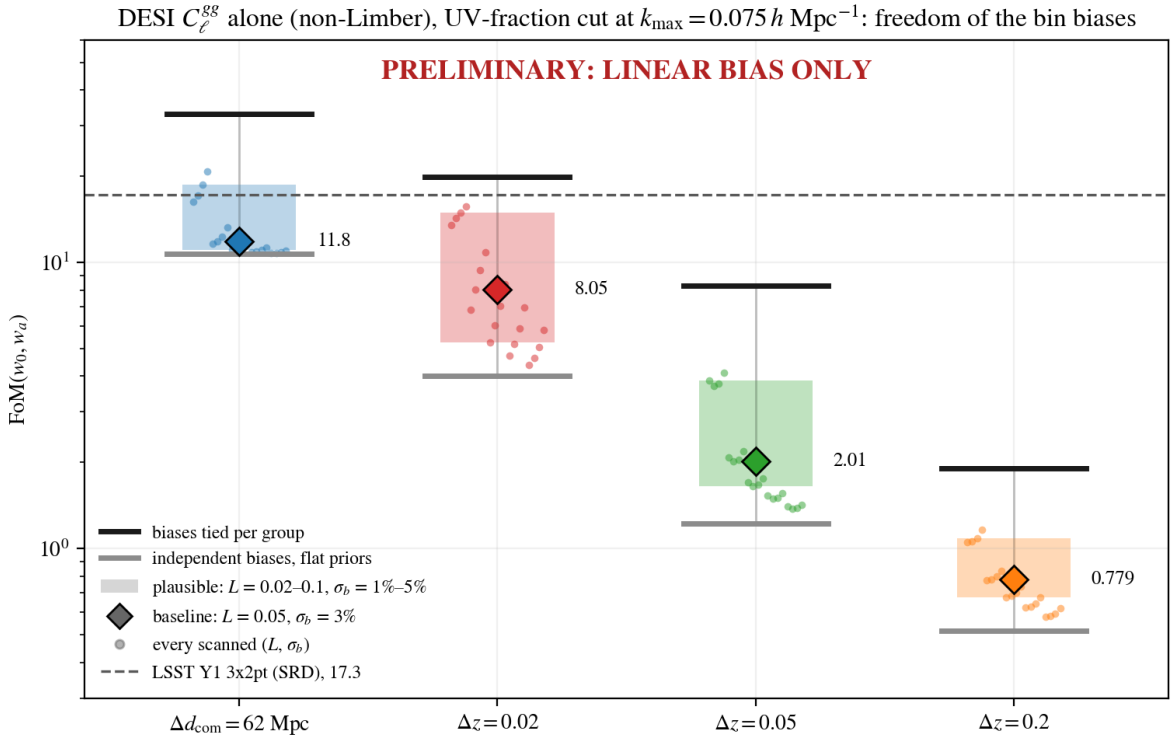


Figure 1: The FoM(w_0, w_a) of each thin-bin layout: the tied model (black cap), independent biases (gray cap), every scanned (L, σ_b) point (dots), the plausible region (σ_b = 1–5%, L = 0.02–0.1; band), the baseline (diamond), and the LSST Y1 3x2pt reference at 17.3 (dashed).

Summary

The intermediate bias model frees one amplitude B_g per tracer group and lets each bin deviate fractionally from the group's constant-amplitude trend $B_g/D(z)$, with deviations correlated over a redshift length L and a per-bin wander σ_b. In the Fisher matrix this becomes one projected precision P per group, added to the bin-amplitude block: P charges for differences between neighboring amplitudes but never for a uniform shift of the group, links only neighboring bins directly (the inverse covariance is tridiagonal, even though the correlation itself reaches far), and reduces to the tied model as $L \to \infty$ and to independent flat-prior biases as $\sigma_b \to \infty$.

A The projected precision for two bins

Everything essential already happens with two bins. Call their amplitudes β_1 and β_2, with common fiducial value B, prior spread s per bin, and correlation coefficient r between the two deviations. The prior covariance and its inverse are

$$\mathsf{C} = s^2 \begin{pmatrix} 1 & r \\ r & 1 \end{pmatrix}, \qquad \mathsf{C}^{-1} = \frac{1}{s^2\,(1-r^2)} \begin{pmatrix} 1 & -r \\ -r & 1 \end{pmatrix}. \tag{16}$$

(The inverse is checked by multiplying: the off-diagonal of $\mathsf{C}\,\mathsf{C}^{-1}$ is $r - r = 0$ and the diagonal is $(1 - r^2)/(1 - r^2) = 1$.)

Now examine the prior penalty $\chi^2_{\text{prior}} = \Delta\beta^{\mathsf{T}}\,\mathsf{C}^{-1}\Delta\beta$, where $\Delta\beta = (\beta_1 - B,\ \beta_2 - B)^{\mathsf{T}}$ is the displacement from the fiducial. Two kinds of move matter.

A uniform shift. Take $\Delta\beta = a\,(1, 1)^{\mathsf{T}}$: both bins move up by the same amount a. Working out the quadratic form with Eq. (16),

$$\Delta\beta^{\mathsf{T}}\mathsf{C}^{-1}\Delta\beta \;=\; \frac{a^2}{s^2(1 - r^2)}\,\big[\,(1 - r) + (1 - r)\,\big] \;=\; \frac{2a^2}{s^2\,(1 + r)} \;\neq\; 0. \tag{17}$$

The plain prior charges for a uniform shift. But a uniform shift of all β_i is a change of B_g, a direction the tied model leaves completely free. If the intermediate model kept this penalty, it would constrain B_g and the comparison between the models would be unfair.

A shape change. Take $\Delta\beta = a\,(1, -1)^{\mathsf{T}}$: the bins move apart. The same algebra gives

$$\Delta\beta^{\mathsf{T}}\mathsf{C}^{-1}\Delta\beta \;=\; \frac{2a^2}{s^2\,(1 - r)}. \tag{18}$$

This is the penalty the prior is *meant* to apply: it grows without bound as $r \to 1$, where the two bins are forced to track each other, and it relaxes to the independent value $2a^2/s^2$ as $r \to 0$.

A number makes the contrast concrete. At $r = 0.670$ (the worked value of Section 2.2), the shape change costs $2a^2/(0.330\,s^2) = 6.1\,a^2/s^2$ while the uniform shift costs $2a^2/(1.670\,s^2) = 1.2\,a^2/s^2$. The correlated prior already charges five times more for the shape change than for the uniform shift; the projection below takes the uniform-shift charge all the way to zero.

The projected precision. The repair for the uniform shift is the projection P of Eq. (12). Appendix B explains where the formula comes from; here, compute it for two bins. Because the two rows of C^{-1} have the same sum, $\mathsf{C}^{-1}\mathbf{1} = \big[s^2(1+r)\big]^{-1}(1, 1)^{\mathsf{T}}$ and $\mathbf{1}^{\mathsf{T}}\mathsf{C}^{-1}\mathbf{1} = 2/\big[s^2(1+r)\big]$. Substituting into Eq. (12) and simplifying each entry,

$$\mathsf{P} \;=\; \frac{1}{2s^2(1 - r)}\begin{pmatrix} 1 & -1 \\ -1 & 1 \end{pmatrix}, \qquad \Delta\beta^{\mathsf{T}}\mathsf{P}\,\Delta\beta \;=\; \frac{(\beta_1 - \beta_2)^2}{2s^2(1 - r)}. \tag{19}$$

The projected prior charges only for the *difference* of the two amplitudes. A uniform shift, $\beta_1 = \beta_2$, is now free, as in the tied model; a shape change keeps exactly the penalty of Eq. (18).

The denominator is not an arbitrary constant. With $\mathrm{Var}(\beta_i) = s^2$ and $\mathrm{Cov}(\beta_1, \beta_2) = s^2 r$,

$$\mathrm{Var}(\beta_1 - \beta_2) \;=\; s^2 + s^2 - 2s^2 r \;=\; 2s^2(1 - r)\,: \tag{20}$$

the projected penalty is the squared jump between the bins divided by the prior variance of that jump. The whole construction reduces to one sentence: *the prior constrains the jump, not the level.*

B Why the projection "sees" the uniform shift: three views

Equation (12) subtracts from C^{-1} a matrix built out of the single vector $v = \mathsf{C}^{-1}\mathbf{1}$: the subtracted piece is $v\,v^{\mathsf{T}}$ divided by the number $\mathbf{1}^{\mathsf{T}}\mathsf{C}^{-1}\mathbf{1}$. A matrix built from one vector acts along one direction only, so the subtraction removes the prior's information about exactly one direction, the uniform shift, and leaves everything else alone. This section derives the formula three times, each derivation adding its own intuition. Throughout, a group has n bins; $\Delta\beta$ and $\mathbf{1}$ are vectors with n entries.

B.1 View (a): subtracting the best-fitting uniform shift

Ask the following question: given a displacement $\Delta\beta$, how much of it can be explained as a pure change of the group amplitude? A change of B_g by c moves every β_i by c, so the candidate explanation is $c\,\mathbf{1}$, and the unexplained part should be judged by the prior metric. Minimize over the scalar c the quadratic form

$$\chi^2(c) \;=\; \big(\Delta\beta - c\,\mathbf{1}\big)^{\mathsf{T}}\mathsf{C}^{-1}\big(\Delta\beta - c\,\mathbf{1}\big). \tag{21}$$

Differentiating with respect to c (the two cross terms are equal because C^{-1} is symmetric),

$$\frac{d\chi^2}{dc} \;=\; -2\,\mathbf{1}^{\mathsf{T}}\mathsf{C}^{-1}\big(\Delta\beta - c\,\mathbf{1}\big) \;=\; 0, \qquad \frac{d^2\chi^2}{dc^2} \;=\; 2\,\mathbf{1}^{\mathsf{T}}\mathsf{C}^{-1}\mathbf{1} \;>\; 0, \tag{22}$$

where the second derivative is positive because C^{-1} is positive definite, so the solution of the first equation is the minimum of $\chi^2(c)$. That solution is

$$a \;=\; \frac{\mathbf{1}^{\mathsf{T}}\mathsf{C}^{-1}\Delta\beta}{\mathbf{1}^{\mathsf{T}}\mathsf{C}^{-1}\mathbf{1}}. \tag{23}$$

For a diagonal C with variances s_i^2 this reduces to $a = \sum_i \Delta\beta_i/s_i^2 \big/ \sum_i 1/s_i^2$, the familiar inverse-variance weighted mean; Eq. (23) is its generalization to correlated errors: generalized least squares for a constant.

Keep the dimensions straight: $\Delta\beta$, $\mathbf{1}$ and the residual $w = \Delta\beta - a\,\mathbf{1}$ are vectors with n entries; a is a single number. The residuals w satisfy one linear constraint (Eq. (25) below), so they span $n-1$ dimensions: one direction of the n-dimensional displacement space has been handed to the amplitude, and $n-1$ directions of genuine shape freedom remain.

B.2 View (b): a Pythagorean split of the penalty

Write the displacement as the best uniform shift plus a residual,

$$\Delta\beta \;=\; a\,\mathbf{1} + w. \tag{24}$$

The first-order condition of Eq. (22) says precisely that

$$\mathbf{1}^{\mathsf{T}}\mathsf{C}^{-1}w \;=\; 0\,: \tag{25}$$

the residual is orthogonal to the uniform direction *in the* C^{-1} *metric*. (With correlated errors, "orthogonal" must be measured with the same matrix that measures lengths.) Substituting the split into the full penalty, the cross term is $2a\,\mathbf{1}^{\mathsf{T}}\mathsf{C}^{-1}w = 0$, and

$$\Delta\beta^{\mathsf{T}}\mathsf{C}^{-1}\Delta\beta \;=\; a^2\,\mathbf{1}^{\mathsf{T}}\mathsf{C}^{-1}\mathbf{1} \;+\; w^{\mathsf{T}}\mathsf{C}^{-1}w. \tag{26}$$

This is the Pythagorean theorem in the prior metric: the squared length of the displacement is the squared length of its uniform piece plus the squared length of its shape piece.

Now compare with the projection. Inserting Eq. (23) into the subtracted term of Eq. (12) shows that the subtraction removes exactly $a^2\,\mathbf{1}^{\mathsf{T}}\mathsf{C}^{-1}\mathbf{1}$, the uniform piece. What remains is

$$\Delta\beta^{\mathsf{T}}\mathsf{P}\,\Delta\beta \;=\; w^{\mathsf{T}}\mathsf{C}^{-1}w. \tag{27}$$

Two consequences follow immediately. First, a pure uniform shift has $w = 0$, so its penalty vanishes. Equivalently $\mathsf{P}\,\mathbf{1} = 0$: the projected matrix sends the uniform direction to zero and charges nothing for it. Second, a pure shape change (a displacement that already satisfies Eq. (25)) has $a = 0$ and keeps its original penalty untouched. The projection removes one direction and alters nothing else.

B.3 View (c): marginalizing a flat-prior amplitude

The third view explains why the projection, rather than some other uniform-shift repair, is the *correct* prior. Model the group amplitudes as

$$\beta \;=\; B\,\mathbf{1} + \varepsilon, \qquad \varepsilon \sim \mathcal{N}(0, \mathsf{C}), \tag{28}$$

with a flat prior on the scalar B. This is literally the intermediate model of Eq. (2), linearized around the fiducial. The prior density of β is obtained by integrating B out:

$$p(\beta) \;\propto\; \int dB\, \exp\!\Big[-\tfrac{1}{2}\big(\beta - B\,\mathbf{1}\big)^{\mathsf{T}}\mathsf{C}^{-1}\big(\beta - B\,\mathbf{1}\big)\Big]. \tag{29}$$

The exponent is quadratic in B. Writing $S = \mathbf{1}^{\mathsf{T}}\mathsf{C}^{-1}\mathbf{1}$ and completing the square,

$$\big(\beta - B\mathbf{1}\big)^{\mathsf{T}}\mathsf{C}^{-1}\big(\beta - B\mathbf{1}\big) \;=\; S\,\big(B - B_*\big)^2 \;+\; \beta^{\mathsf{T}}\mathsf{C}^{-1}\beta \;-\; \frac{\big(\mathbf{1}^{\mathsf{T}}\mathsf{C}^{-1}\beta\big)^2}{S}, \qquad B_* = \frac{\mathbf{1}^{\mathsf{T}}\mathsf{C}^{-1}\beta}{S}. \tag{30}$$

The B integral is a Gaussian centered on B_* with width $1/\sqrt{S}$; it contributes only a β-independent constant $\sqrt{2\pi/S}$. What survives in the exponent is

$$p(\beta) \;\propto\; \exp\!\Big[-\tfrac{1}{2}\,\beta^{\mathsf{T}}\mathsf{P}\,\beta\Big], \qquad \mathsf{P} \;=\; \mathsf{C}^{-1} - \frac{\mathsf{C}^{-1}\mathbf{1}\,\mathbf{1}^{\mathsf{T}}\mathsf{C}^{-1}}{\mathbf{1}^{\mathsf{T}}\mathsf{C}^{-1}\mathbf{1}}, \tag{31}$$

exactly the projection of Eq. (12).

The same computation has a standard matrix name. Expanding the exponent of Eq. (29) in the joint variables (β, B) gives a Gaussian with the $(n+1)\times(n+1)$ block precision

$$\mathsf{F}_{(\beta,B)} \;=\; \begin{pmatrix} \mathsf{C}^{-1} & -\,\mathsf{C}^{-1}\mathbf{1} \\ -\,\mathbf{1}^{\mathsf{T}}\mathsf{C}^{-1} & \mathbf{1}^{\mathsf{T}}\mathsf{C}^{-1}\mathbf{1} \end{pmatrix}. \tag{32}$$

Removing a parameter from a precision matrix is not just deleting its row and column: the remaining block must also be corrected downward by whatever the removed parameter could absorb. The corrected block, $\mathsf{P} = \mathsf{F}_{\beta\beta} - \mathsf{F}_{\beta B}\,\mathsf{F}_{BB}^{-1}\,\mathsf{F}_{B\beta}$, is called the *Schur complement* over B, and it reproduces Eq. (12) term by term. So P is not a trick: it is precisely the statement "B_g *is sampled with a flat prior and marginalized*", folded into one matrix.

Note also that B_* in Eq. (30) is the same generalized-least-squares mean as a in Eq. (23): the marginalization integrates over all amplitudes, and the best-fitting one dominates the integral.

B.4 Checking views (b) and (c) on the two-bin case

For $n = 2$ the rows of C^{-1} in Eq. (16) have equal sums, so Eq. (23) gives the plain mean, $a = (\Delta\beta_1 + \Delta\beta_2)/2$, and the residual is the antisymmetric half-difference, $w = \frac{1}{2}(\Delta\beta_1 - \Delta\beta_2)\,(1,-1)^{\mathsf{T}}$. Its penalty, using the shape-change result of Eq. (18) with $a \to \frac{1}{2}(\Delta\beta_1 - \Delta\beta_2)$, is

$$w^{\mathsf{T}}\mathsf{C}^{-1}w \;=\; \frac{2}{s^2(1-r)}\left(\frac{\Delta\beta_1 - \Delta\beta_2}{2}\right)^2 \;=\; \frac{(\Delta\beta_1 - \Delta\beta_2)^2}{2s^2(1-r)}, \tag{33}$$

which is the direct two-bin computation of Eq. (19). Views (a), (b) and (c) all land on the same matrix.

C Many bins and several groups

C.1 Three bins: total correlation versus direct links

A group now has n bins at redshifts $z_1 < z_2 < \cdots < z_n$, unevenly spaced in general, with the prior covariance of Eq. (5),

$$\mathsf{C}_{ij} \;=\; s^2 \exp\!\left(-\frac{|z_i - z_j|}{L}\right). \tag{34}$$

This matrix is positive definite for any set of distinct redshifts (all its eigenvalues are positive), so the prior is a proper probability distribution and C^{-1} exists.

The structure of C^{-1} is best seen on three evenly spaced bins. With neighbor correlation $r = e^{-\Delta z/L}$, the outer pair sits two steps apart and carries $e^{-2\Delta z/L} = r^2$, so

$$\mathsf{C} \;=\; s^2\begin{pmatrix} 1 & r & r^2 \\ r & 1 & r \\ r^2 & r & 1 \end{pmatrix}, \qquad \mathsf{C}^{-1} \;=\; \frac{1}{s^2\,(1-r^2)}\begin{pmatrix} 1 & -r & 0 \\ -r & 1+r^2 & -r \\ 0 & -r & 1 \end{pmatrix}. \tag{35}$$

One multiplication confirms the surprising entry. The $(1,3)$ entry of $\mathsf{C}^{-1}\mathsf{C}$ must be zero (it is an off-diagonal entry of the identity matrix), and indeed row 1 of C^{-1} times column 3 of C gives, dropping the prefactors,

$$1\cdot r^2 \;+\; (-r)\cdot r \;+\; 0\cdot 1 \;=\; r^2 - r^2 \;=\; 0. \tag{36}$$

Here is the point that looks paradoxical at first. Bins 1 and 3 *are* correlated (C contains r^2), yet their entry in C^{-1} is exactly zero. The two matrices answer different questions. C gives the *total* correlation: what δ_1 tells you about δ_3 when nothing else is known. C^{-1} gives the *direct* correlation: what δ_1 still tells you about δ_3 once δ_2 is already known, and that is nothing. All of the 1–3 correlation travels through bin 2.

Think of a chain of people passing a message. The first person's message reaches far down the chain, but the links are only between neighbors: nobody talks past the person standing next to them.

The exponential kernel is special in making this exact, because its correlations multiply along the chain:

$$e^{-(z_3 - z_1)/L} \;=\; e^{-(z_3 - z_2)/L}\, e^{-(z_2 - z_1)/L}. \tag{37}$$

The long-distance correlation is the product of the step correlations, so the middle bin already carries everything the outer bins share. A Gaussian kernel $e^{-\Delta z^2/L^2}$ does not factorize this way,

because the squared separations do not add up along the chain: $(z_3-z_1)^2 \neq (z_3-z_2)^2+(z_2-z_1)^2$. Its inverse is dense.

The same structure shows directly in the penalty. For n evenly spaced bins, the quadratic form of the deviations δ_i (whose covariance is $\sigma_b^2\mathsf{R}$) regroups into a sum over neighbor links,

$$\delta^{\mathsf{T}}\big(\sigma_b^2\mathsf{R}\big)^{-1}\delta \;=\; \frac{\delta_1^2}{\sigma_b^2} \;+\; \sum_{k=1}^{n-1}\frac{\big(\delta_{k+1}-r\,\delta_k\big)^2}{\sigma_b^2\,(1-r^2)}. \tag{38}$$

Each term compares one bin with r times the bin before it: every bin is pulled only by its neighbor, and the long-range correlation comes from chaining the pulls. (Expanding the squares and collecting terms reproduces Eq. (35) for $n = 3$; Appendix D does it for general, uneven spacing.) This is the precise sense of "correlation only between adjacent bins": *directly*, only neighbors talk; *in total*, the influence propagates down the whole chain. Probability theory calls this neighbors-only structure the Markov property.

C.2 The tridiagonal inverse for n bins

A matrix whose nonzero entries sit only on the diagonal and the two neighboring off-diagonals is called *tridiagonal*. The three-bin inverse of Eq. (35) is the smallest interesting example, and the pattern grows unchanged. For n evenly spaced bins, where every adjacent pair has the same correlation $r = e^{-\Delta z/L}$ and $\mathsf{C}_{ij} = s^2\,r^{|i-j|}$,

$$\mathsf{C}^{-1} \;=\; \frac{1}{s^2\,(1-r^2)}\begin{pmatrix} 1 & -r & & & \\ -r & 1+r^2 & -r & & \\ & -r & \ddots & \ddots & \\ & & \ddots & 1+r^2 & -r \\ & & & -r & 1 \end{pmatrix}, \tag{39}$$

with diagonal $(1,\ 1+r^2,\ \ldots,\ 1+r^2,\ 1)$, off-diagonal entries $-r$, blanks zero, and the overall factor $1/\big[s^2(1-r^2)\big]$. The entries read off the chain sum of Eq. (38). An off-diagonal entry is the link between that pair of neighbors. An interior bin touches two links, one to the bin on its left and one to the bin on its right, and the two pulls together contribute $1+r^2$; the first and last bins touch only one link each, hence the plain 1 in the corners.

With uneven spacing, each adjacent pair has its own step and therefore its own correlation,

$$r_k \;=\; \exp\!\left(-\,\frac{z_{k+1}-z_k}{L}\right), \qquad k = 1,\ldots,n-1, \tag{40}$$

and the inverse keeps the same shape with per-pair entries:

$$s^2\left(\mathsf{C}^{-1}\right)_{kk} = \begin{cases} \dfrac{1}{1-r_1^2}, & k=1, \\[2ex] \dfrac{1}{1-r_{k-1}^2}+\dfrac{r_k^2}{1-r_k^2}, & 1<k<n, \\[2ex] \dfrac{1}{1-r_{n-1}^2}, & k=n, \end{cases} \qquad s^2\left(\mathsf{C}^{-1}\right)_{k,k+1} = \frac{-\,r_k}{1-r_k^2}. \tag{41}$$

In words: the off-diagonal entry is the neighbor link of pair k, stronger when the bins are closer (r_k nearer one); the diagonal entry of bin k collects the links touching that bin, one term from the pair on its left and one from the pair on its right; the first and last bins carry one link only. Setting all $r_k = r$ turns the interior diagonal into $1/(1-r^2) + r^2/(1-r^2) = (1+r^2)/(1-r^2)$ and recovers Eq. (39). Appendix D derives Eq. (41) from the chain of neighbor links; the companion script `verify_prior.py` checks the three-bin, even and uneven forms numerically.

C.3 The projection, group by group

Each group receives the projected precision of Eq. (12), built from its own C. The properties derived in Appendix B carry over verbatim to n bins. $\mathsf{P}\,\mathbf{1} = 0$: a uniform shift of the whole group costs nothing. P is positive semidefinite, which means no displacement is *rewarded*: every move either costs χ^2 or is free. And of the n directions a group can move in, exactly one (the uniform shift) is free while the other $n-1$ are charged: in matrix language, P has rank $n-1$.

Several groups combine trivially. The prior of Eq. (3) is zero across groups, so the full prior covariance is block-diagonal, one block per group; and each group has its own free amplitude B_g, so each block is projected separately. The total prior precision is the block-diagonal matrix of the per-group projections.

C.4 The two limits

The intermediate model contains both extremes.

As $L \to \infty$, every $r_k \to 1$, so the jump variance $2\sigma_b^2(1-r_k)$ of every adjacent pair goes to zero: the prior forces all deviations of a group to be equal. The only freedom left is their common value, a uniform shift, and that is the direction the projection leaves free. This is the tied model, with one free B_g per group, *whatever* the value of σ_b.

As $\sigma_b \to \infty$ at fixed L, the overall scale of C diverges, so $\mathsf{C}^{-1} \to 0$ and with it $\mathsf{P} \to 0$: the prior penalty vanishes and every bin's amplitude is free: the independent model with flat priors.

Between the limits, L and σ_b interpolate continuously, which is what makes the model useful: the figure of merit can be traced as a function of how much bin-to-bin freedom the bias is granted.

D The tridiagonal inverse from the chain of links

Because correlations multiply along the chain (Eq. (37)), the prior can be built one bin at a time: the joint density of the deviations factorizes into neighbor conditionals,

$$p(\delta_1, \ldots, \delta_n) \;=\; p(\delta_1)\, p(\delta_2 \mid \delta_1) \cdots p(\delta_n \mid \delta_{n-1}), \tag{42}$$

where each conditional is Gaussian with

$$\big\langle \delta_{k+1} \mid \delta_k \big\rangle \;=\; r_k\, \delta_k, \qquad \mathrm{Var}\big(\delta_{k+1} \mid \delta_k\big) \;=\; \sigma_b^2\,\big(1 - r_k^2\big). \tag{43}$$

Here $r_k = e^{-(z_{k+1}-z_k)/L}$ is, as in Sections 2.2 and C.2, the correlation coefficient of the adjacent bins k and $k{+}1$: the kernel of Eq. (3) gives $\mathrm{Cov}(\delta_k, \delta_{k+1}) = \sigma_b^2\, r_k$ and $\mathrm{Var}(\delta_k) = \mathrm{Var}(\delta_{k+1}) = \sigma_b^2$.

The conditional mean and variance in Eq. (43) follow from nothing more than these covariances. Split δ_{k+1} into the part predicted by its neighbor and a leftover e,

$$\delta_{k+1} \;=\; a\,\delta_k + e, \tag{44}$$

and choose the number a so that the leftover carries no correlation with δ_k:

$$\mathrm{Cov}(e,\delta_k) \;=\; \mathrm{Cov}(\delta_{k+1},\delta_k) - a\,\mathrm{Var}(\delta_k) \;=\; \sigma_b^2\, r_k - a\,\sigma_b^2 \;=\; 0 \quad\Longrightarrow\quad a = r_k. \tag{45}$$

The leftover's variance follows by expanding $e = \delta_{k+1} - r_k\delta_k$:

$$\mathrm{Var}(e) \;=\; \mathrm{Var}(\delta_{k+1}) - 2r_k\,\mathrm{Cov}(\delta_{k+1},\delta_k) + r_k^2\,\mathrm{Var}(\delta_k) \;=\; \sigma_b^2 - 2r_k^2\sigma_b^2 + r_k^2\sigma_b^2 \;=\; \sigma_b^2\,\big(1-r_k^2\big). \tag{46}$$

For jointly Gaussian variables, uncorrelated means independent. So once δ_k is known, the only randomness left in $\delta_{k+1} = r_k\delta_k + e$ is e itself: the conditional mean is $r_k\delta_k$ and the conditional variance is $\sigma_b^2(1-r_k^2)$, which is Eq. (43). In one line: knowing the neighbor explains the fraction r_k^2 of the bin's variance, and the fraction $1 - r_k^2$ is new.

The factorization of Eq. (42) needs one more fact: conditioning on δ_k alone must be enough: the earlier bins must add nothing. This is again the multiplication of correlations along the chain. For any earlier bin $j \le k$,

$$\mathrm{Cov}(e,\delta_j) \;=\; \sigma_b^2\left[e^{-(z_{k+1}-z_j)/L} - r_k\,e^{-(z_k-z_j)/L}\right] \;=\; \sigma_b^2\,e^{-(z_k-z_j)/L}\left[e^{-(z_{k+1}-z_k)/L} - r_k\right] \;=\; 0, \tag{47}$$

because the first exponential factorizes through z_k and the bracket is $r_k - r_k$. The leftover is therefore independent of every earlier bin, and the chain of one-step conditionals is the complete prior.

Taking $-2\ln p$ of Eq. (42) gives the penalty as the chain sum, the uneven-spacing version of Eq. (38):

$$\delta^{\mathsf{T}}\big(\sigma_b^2\mathsf{R}\big)^{-1}\delta \;=\; \frac{\delta_1^2}{\sigma_b^2} \;+\; \sum_{k=1}^{n-1}\frac{\big(\delta_{k+1} - r_k\,\delta_k\big)^2}{\sigma_b^2\,\big(1-r_k^2\big)}. \tag{48}$$

Expanding each square and collecting the coefficient of every product $\delta_i\delta_j$ reads off the precision matrix entry by entry. δ_1^2 appears twice: in the first term (coefficient 1) and in the $k=1$ link as the predictor (coefficient $r_1^2/(1-r_1^2)$); the two add up to $1/(1-r_1^2)$. An interior δ_k^2 appears in the $k-1$ link as the predicted bin ($1/(1-r_{k-1}^2)$) and in the k link as the predictor ($r_k^2/(1-r_k^2)$). The last bin is only ever predicted: $1/(1-r_{n-1}^2)$. Each product $\delta_k\delta_{k+1}$ appears in its own link only, with coefficient $-2r_k/(1-r_k^2)$, i.e. a symmetric matrix entry $-r_k/(1-r_k^2)$. No product of non-neighbors ever appears, so the matrix is tridiagonal by construction. This is Eq. (41) in units of σ_b^2; replacing δ by $\Delta\beta = B_g\,\delta$ and σ_b by $s = \sigma_b B_g$ gives the same matrix for the bin amplitudes.

E The bias continuity equation, step by step

The matter density $\rho_m(\mathbf{x},t)$ defines the matter overdensity $\delta_m = (\rho_m - \bar\rho_m)/\bar\rho_m$, with $\bar\rho_m(t)$ the mean density; the galaxy number density n_g defines the galaxy-count overdensity δ_g the same way. Conservation of mass and of galaxy number gives each field a continuity equation. In

comoving coordinates, keeping only terms linear in the fluctuations (the product $\delta\,\mathbf{v}$ is second order and dropped),

$$\frac{d\delta_m}{dt} \;=\; -\,\frac{1}{a}\,\nabla\cdot\mathbf{v}_m, \qquad \frac{d\delta_g}{dt} \;=\; -\,\frac{1}{a}\,\nabla\cdot\mathbf{v}_g, \tag{49}$$

where t is cosmic time, a the scale factor, and $\mathbf{v}_m$ and $\mathbf{v}_g$ the peculiar velocity fields of the matter and of the galaxies.

The two velocity fields are the same. On large scales the galaxies carry no pressure of their own and fall in the same gravitational potential as the matter, so a conserved sample rides the matter flow: $\mathbf{v}_g = \mathbf{v}_m$. This is the no-velocity-bias assumption; Tegmark and Peebles [2] state it as galaxies behaving as test particles that fairly trace the large-scale velocity field.

With equal right-hand sides, subtracting the two equations of Eq. (49) leaves

$$\frac{d}{dt}\big(\delta_g - \delta_m\big) \;=\; 0 \quad\Longrightarrow\quad \delta_g(z) - \delta_m(z) \;=\; \delta_g(z_c) - \delta_m(z_c) \tag{50}$$

after integrating in time from the reference redshift z_c to any other redshift z: the excess clumping keeps its initial value forever.

Now substitute the bias. At z_c the sample has linear bias b_c, so $\delta_g(z_c) = b_c\,\delta_m(z_c)$ and the frozen difference is $(b_c-1)\,\delta_m(z_c)$. The matter grows as $\delta_m(z) = \delta_m(z_c)\,D(z)/D(z_c)$, with D the linear growth factor. Then

$$\delta_g(z) \;=\; \delta_m(z) + \big(b_c-1\big)\,\delta_m(z_c) \;=\; \delta_m(z)\left[\,1 + \big(b_c-1\big)\,\frac{D(z_c)}{D(z)}\,\right], \tag{51}$$

and dividing by $\delta_m(z)$ gives the bias $b(z) = \delta_g/\delta_m$ of Eq. (13). Dividing instead by the tied bias $b_c\,D(z_c)/D(z)$ gives the deviation of Eq. (14), $\delta(z) = (D(z)/D(z_c)-1)/b_c$: positive at redshifts below z_c, where the matter has grown past its reference value, and negative above.

The debiasing limit follows from Eq. (13) alone. As D grows, the frozen term is divided by an ever larger number, so $b \to 1$: from above when $b_c > 1$ and from below when $b_c < 1$. Gravity debiases a conserved sample, because the part of δ_g that traces the matter keeps growing while the frozen excess does not.

Tegmark and Peebles drop the assumption that δ_g is proportional to δ_m point by point. They describe the pair of fields by three numbers: the matter fluctuation amplitude σ, the bias b (the ratio of the galaxy to the matter fluctuation amplitude) and the correlation coefficient r between the two fields; $r < 1$ is stochastic bias, galaxies tracing the matter imperfectly. The same continuity argument evolves all three (their section "After the galaxy formation epoch"): with D normalized to one today and the subscript zero marking today,

$$b_0 = \big[(1-D)^2 + 2D(1-D)\,b\,r + D^2 b^2\big]^{1/2}, \qquad r_0 = \frac{(1-D) + D\,b\,r}{b_0}, \tag{52}$$

where b and r are the values at the epoch whose growth factor is D.

Both b_0 and r_0 approach one as clustering grows, whatever their initial values: the matter-sourced part of the galaxy field grows with D while the frozen part does not, so the field is progressively dominated by the piece that traces the matter. Perfect correlation is a fixed point: for $r = 1$, Eq. (52) collapses to $b_0 - 1 = (b-1)\,D$, which is Eq. (13) again. The conserved population of this note is the perfectly correlated special case of their model.

Each assumption fails in a specific way for a DESI sample. Membership is not conserved: the flux limit and color cuts admit and remove galaxies as redshift grows (the threshold drift

and color-selection features of Section 3.4), mergers turn two members into one, and an ELG belongs to the sample only during its star-forming phase. The no-velocity-bias and linear-theory assumptions hold well on the scales this forecast keeps; conservation is the assumption that breaks.

The script of Appendix H integrates Eq. (49) numerically, with δ_m from the linear growth equation, and recovers the closed form of Eq. (13) to a relative accuracy of 2×10^{-9}.

F What the literature says, paper by paper

This appendix summarizes each measurement behind Section 3.5, for a reader who has not read the papers. The tags are those of that section: measured, derived, assumed.

F.1 Passive-evolution tests of massive red galaxies

Wake et al. [3]. What was measured: the two-point correlation function of over 10,000 luminous red galaxies (LRGs) from SDSS and the 2SLAQ survey at $0.19 < z < 0.55$, matched in rest-frame color and luminosity, with stellar masses above $10^{11}\,h^{-1}M_\odot$. Method: the redshift-space and real-space autocorrelation of the matched samples, interpreted with a halo model. Result: no significant evolution of the correlation amplitude, but a strictly merger-free passive model is rejected; a merger rate of $7.5 \pm 2.3\%$ between $z = 0.55$ and $z = 0.19$, about 2.4% per Gyr, fits the data. Implication here: LRG deviations from conserved evolution are a few percent in number per Gyr (measured), and the induced bias change is smaller, because merging moves galaxies between halos of similar mass.

Tojeiro et al. [4]. What was measured: the merger rate and clustering evolution of 250,000 BOSS galaxies up to $z \approx 0.7$, connected to the SDSS-I/II LRGs at lower redshift. Method: each BOSS galaxy receives a likelihood of being a progenitor of the LRG sample, built from the fossil record of the LRGs (star-formation and metallicity histories); the weighted sample's large-scale clustering is compared between the surveys. Result: the sample evolves by merging at $2 \pm 1.5\%$ per Gyr, and the weighted clustering evolution is consistent with dynamical passive evolution, the continuity-equation prediction. Implication here: the conserved shape holds to the percent-per-Gyr level for these galaxies (measured).

Guo et al. [5]. What was measured: the projected two-point correlation function of about 260,000 CMASS galaxies at $0.43 < z < 0.7$ (BOSS DR9). Method: subsamples of fixed comoving number density built with redshift-dependent magnitude thresholds, so that the same population is followed across redshift. Result: the clustering of the fixed-density samples is consistent with passive evolution, and red-sequence clustering at fixed luminosity shows no significant evolution. Implication here: the same (measured).

F.2 The DESI luminous red galaxies

Zhou et al. [6]. What was measured: the redshift-dependent clustering of 2.7 million DESI-like LRGs over 5655 square degrees at $0.4 < z < 0.9$, in photometric-redshift bins of width $\Delta z \approx 0.1$. Method: projected correlation functions with photometric redshifts of 0.02 precision, fit with a five-parameter halo occupation distribution (HOD). Result: the HOD

parameters show little evolution except in the highest bin, and the bias follows a constant clustering amplitude; Table 3 lists the measured biases and their residuals around $b = 1.5/D$ (derived here): within $\pm 2.3\%$ below $z = 0.83$, and $+6.4 \pm 6.4\%$ at $0.84 < z < 0.93$. Implication: LRG deviations at $\Delta z \approx 0.1$ spacing are at most a few percent below $z = 0.83$ (measured plus derived).

photometric bin	b	residual around $1.5/D$
0.41–0.50	1.89 ± 0.03	$-0.4\% \pm 1.6\%$
0.50–0.61	2.04 ± 0.02	$+2.3\% \pm 1.0\%$
0.61–0.72	2.07 ± 0.03	$-1.6\% \pm 1.4\%$
0.75–0.83	2.24 ± 0.08	$+0.4\% \pm 3.4\%$
0.84–0.93	2.48 ± 0.15	$+6.4\% \pm 6.4\%$

Table 3: The DESI-like LRG bias of Zhou et al. (2021, arXiv:2001.06018, Results section; measured) and its residual around the constant clustering amplitude $b = 1.5/D$ (derived, with flat ΛCDM $\Omega_m = 0.3$ growth factors at the bin centers).

Zhou et al. [7]. What was measured: the DESI LRG target selection and its spectroscopic performance. Method: selection cuts in g, r, z and $W1$ (the WISE 3.4 micron band), validated on Survey Validation spectra. Result: a sliding color-magnitude cut holds the comoving density near $5 \times 10^{-4}\,(h/\mathrm{Mpc})^3$ at $0.4 < z < 0.8$; the overall faint limit is $z_{\mathrm{fiber}} < 21.6$, where z_{fiber} is the z-band magnitude expected within one DESI fiber; 98.9% of observed targets yield confident redshifts. Implication: inside LRG1 and LRG2 the selection gives the threshold no room to jump (measured); past $z \approx 0.8$ the fixed flux limit drives the LRG3 drift of Section 3.5 (derived).

White et al. [9] and Sailer et al. [10]. What was measured: cosmological constraints from the cross-correlation of the photometric LRG sample of [8] with CMB lensing from Planck, and in [10] also ACT DR6, in four tomographic bins over $0.4 < z < 1$. Method: angular cross- and auto-spectra with one free bias parameter per bin. Result: smooth models fit all bins. Implication here: these analyses neither require nor exclude percent-level bias structure between thin bins; they bound nothing below their bin width of about 0.15 (measured context).

F.3 Bias versus luminosity threshold

Zehavi et al. [12]. What was measured: the luminosity and color dependence of galaxy clustering in the SDSS DR7 main sample, about 700,000 galaxies at $z < 0.25$. Method: projected correlation functions of volume-limited luminosity-threshold samples, interpreted with HOD modeling. Result: the large-scale bias of a threshold sample is fit by $b\,(\sigma_8/0.8) = 1.06 + 0.21\,(L/L_*)^{1.12}$, with L_* the characteristic luminosity of the r-band luminosity function and σ_8 the rms matter fluctuation in spheres of $8\,\mathrm{Mpc}/h$; the measured threshold biases rise from 1.40 ± 0.03 at $M_r < -21.0$ to 2.16 ± 0.05 at $M_r < -22.0$, a slope of 0.5–1.0 in bias per magnitude of threshold near LRG luminosities. Implication: this relation converts a selection-threshold drift into a bias drift; the LRG3 and BGS drifts of Section 3.5 are derived through it.

F.4 The bright galaxy survey

Hahn et al. [13]. What was measured: the BGS target selection and its validation. Method: selection on Legacy Surveys DR9 imaging plus Survey Validation spectra. Result: BGS Bright is a magnitude-limited sample at $r < 19.5$ with redshift success above 95% across a broad range of galaxies; BGS Faint adds $19.5 < r < 20.175$ with a color selection. Implication: the fixed $r < 19.5$ limit raises the luminosity threshold with redshift, which drives the derived 10–25% bias drift across the BGS group (measured limit, derived drift).

Sailer et al. [11]. What was measured: the growth of matter fluctuations at $z \lesssim 0.4$ from BGS imaging cross-correlated with Planck and ACT CMB lensing. Method: tomographic angular cross-spectra with a hybrid effective-field-theory model and per-bin bias nuisance parameters. Result: growth consistent with ΛCDM. Implication here: as with the LRG tomography, no resolution of bias structure between thin bins (measured context).

F.5 The emission-line galaxies

Raichoor et al. [14]. What was measured: the ELG target selection, $0.6 < z < 1.6$, and its spectroscopic efficiency. Method: a g-band fiber magnitude cut plus a $(g-r)$ versus $(r-z)$ color box; redshift reliability is a cut on the signal-to-noise of the [OII] 3727 Å doublet. Result: the [OII] signal-to-noise drops at the specific redshifts where the doublet lands on sky emission lines, especially at $z > 1.5$, and the catastrophic-failure fraction rises over $1.3 < z < 1.6$. Implication: these windows, each $\Delta z \approx 0.001$–0.01 wide, are the one short-separation selection mechanism found in any tracer; their bias amplitude is assumed, of order 1–2%, bounded by the weak dependence of ELG bias on [OII] flux.

Rocher et al. [15]. What was measured: the HOD of the DESI One-Percent ELG sample. Method: fits of AbacusSummit simulation halos to the measured small-scale clustering, in two redshift bins, $0.8 < z < 1.1$ and $1.1 < z < 1.6$. Result: every HOD parameter agrees within one sigma between the bins; the mean host halo mass is $\log_{10} M_h = 11.78^{+0.03}_{-0.04}$ at mean redshift 0.95 and 11.86 ± 0.05 at 1.325. Implication: the ELG bias shape is stable across ELG1 and most of ELG2 (measured); at these host masses the halo bias depends weakly on mass, so the implied drift is at the percent level (derived, with the slope assumed).

G How the prior enters the Fisher matrix

A Gaussian prior adds its precision to the Fisher matrix:

$$F \;=\; F_{\rm data} \;+\; F_{\rm priors}, \tag{53}$$

where $F_{\rm data}$ is the usual data Fisher matrix and $F_{\rm priors}$ holds one block per prior. The projected precision P of each group is added to the rows and columns of that group's bin amplitudes β_i. All other parameters (the cosmological parameters, and the velocity dispersions with their own independent Gaussian priors) are unchanged: P has no entries outside the bias block.

One consequence matters in practice. Bins that lost all their band powers to the scale cut carry no data, so their rows of $F_{\rm data}$ are zero. In the independent model such a parameter is completely unconstrained and must be dropped from the matrix before inversion. With the

correlated prior it stays: P ties the empty bin to its neighbors, which do carry data, so the matrix remains invertible and the empty bin follows the chain.

H Numerical verification

The companion script `verify_prior.py`, in this folder, checks every identity used above: the two-bin closed forms of Eqs. (16) and (19); the three-bin inverse of Eq. (35), including its exactly zero $(1, 3)$ entry; the null direction, positive semidefiniteness and rank $n - 1$ of P on uneven redshifts; the tridiagonal inverses of Eqs. (39) and (41); the Pythagorean identity of Eq. (27) on random displacements; the equality of P with the Schur complement of the joint (β, B) precision of Eq. (32); the decorrelation and variance of the neighbor-prediction leftover of Appendix D; the jump variance of Eq. (8) by Monte Carlo (10^5 draws); the worked example of Section 2.2; the closed-form bias evolution of Eq. (13) against a direct integration of the continuity equation; and the tilt and neighbor-change tables of Section 3. Each check prints a PASS or FAIL line with its measured deviation.

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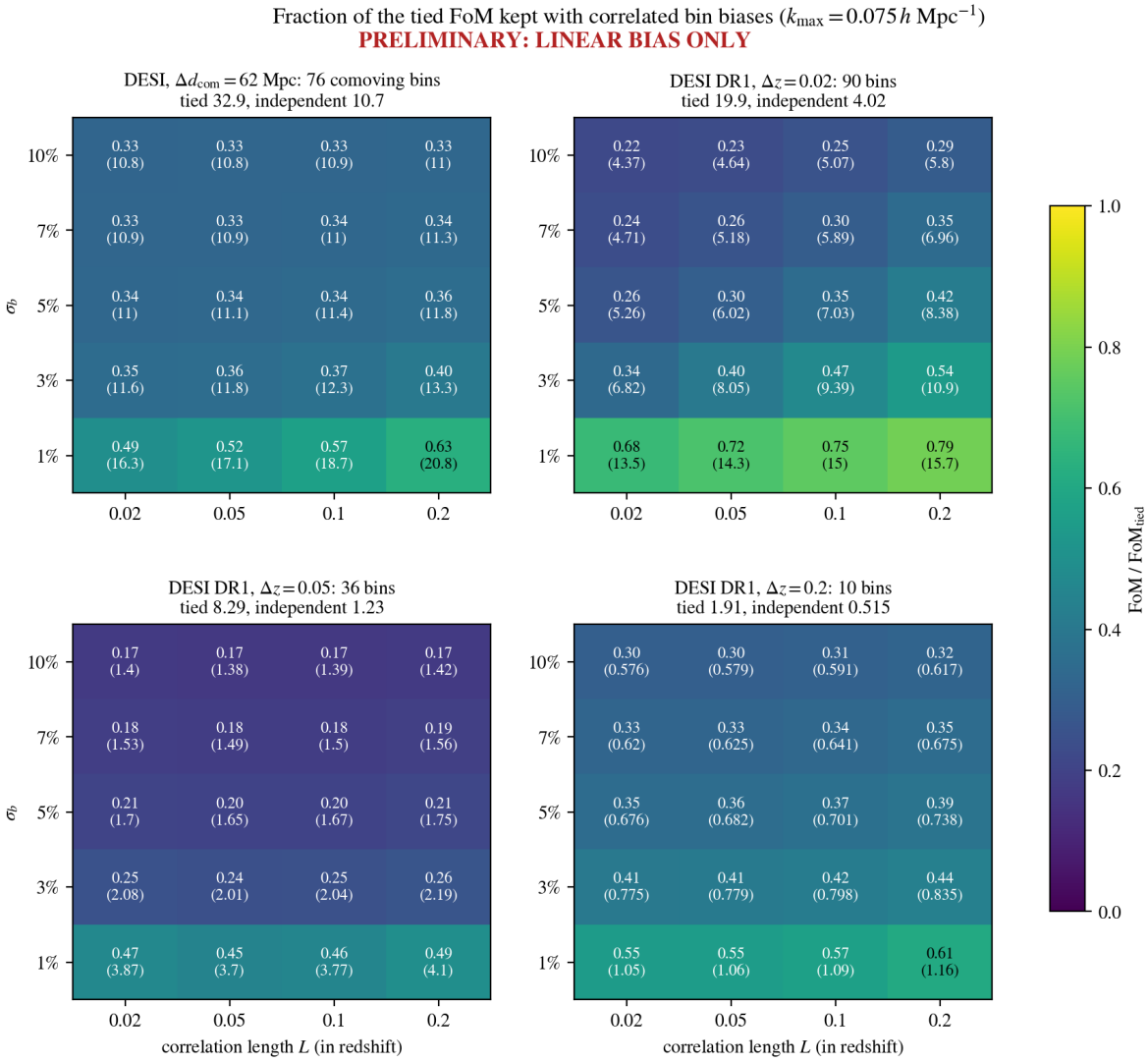


Figure 2: FoM/FoM$_{\text{tied}}$ across the scanned (L, σ_b) grid, one panel per layout, with the absolute FoM in parentheses.

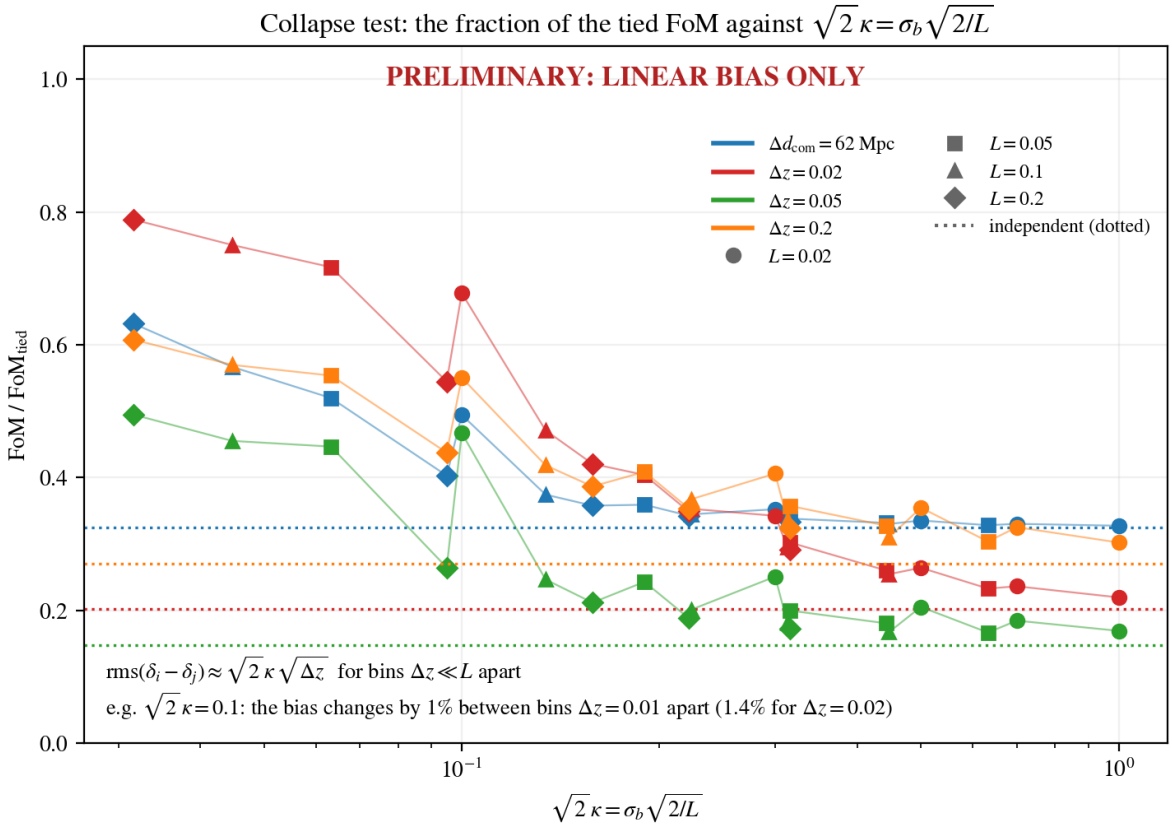


Figure 3: FoM/FoM$_{\rm tied}$ against the roughness combination $\sqrt{2}\,\kappa = \sigma_b\sqrt{2/L}$. Dotted lines: each layout's independent fraction FoM$_{\rm indep}$/FoM$_{\rm tied}$. The points do not collapse onto a single curve per layout.